UTM TENSILE REPORT — UTM_Test_20260819_113054

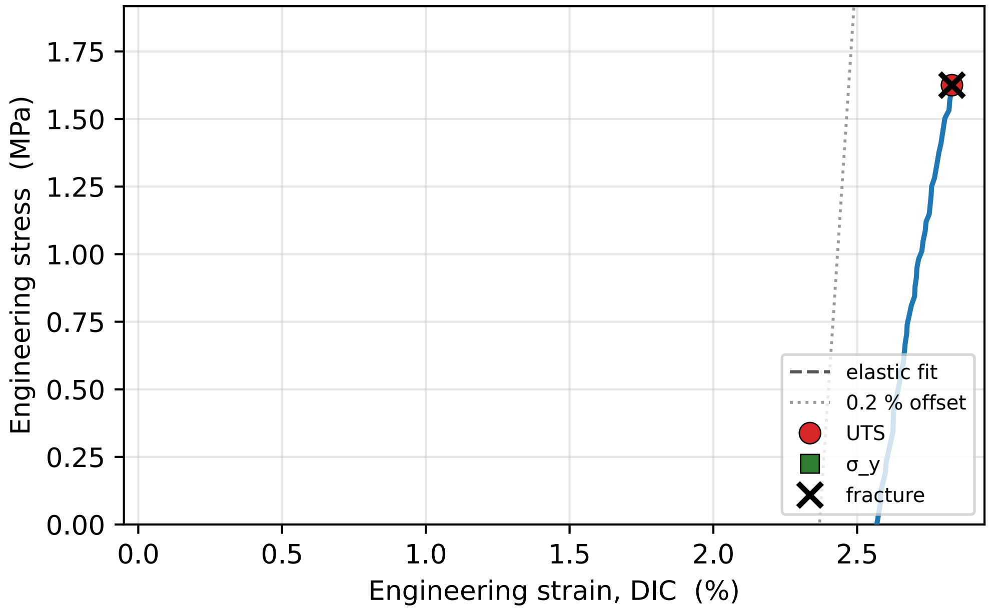
DIC gauge strain (ENGINEERING: $\Delta L/L_0$) · engineering stress (F/A_0) · anchor self-calibration

1.6 MPa	-3.8 MPa	1.60 GPa	2.8 %	-436 kJ/m³	-2791 N
UTS	Yield σ_y	Modulus E	Failure ε_f	Toughness	Preload anchor

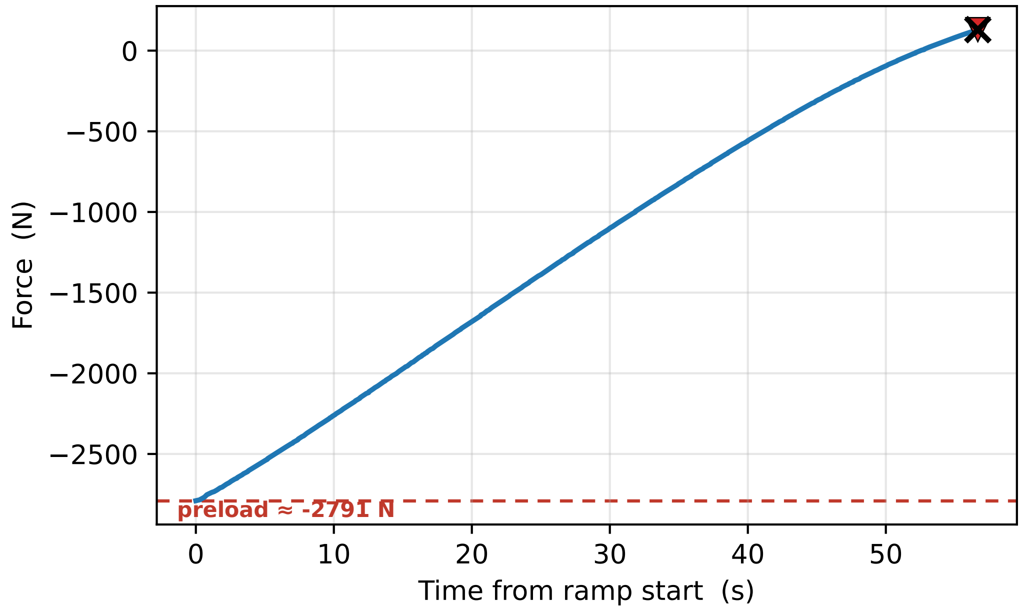
TEST SETTINGS

Specimen	**UTM_Test_20260819_113054**
DIC mode	**Black**
Preload target	**300 N**
Test speed	**0.100 mm/s**
Cross-section	**80 mm²**
Gauge length L_0	**80 mm**
Infill (label)	**100 %**
Force calib	**sc -0.0065 / off 337.077**
Comment	**—**
Test date	**2026-08-19 11:29:26**
Duration	**95.4 s**

Stress–strain



Load vs time

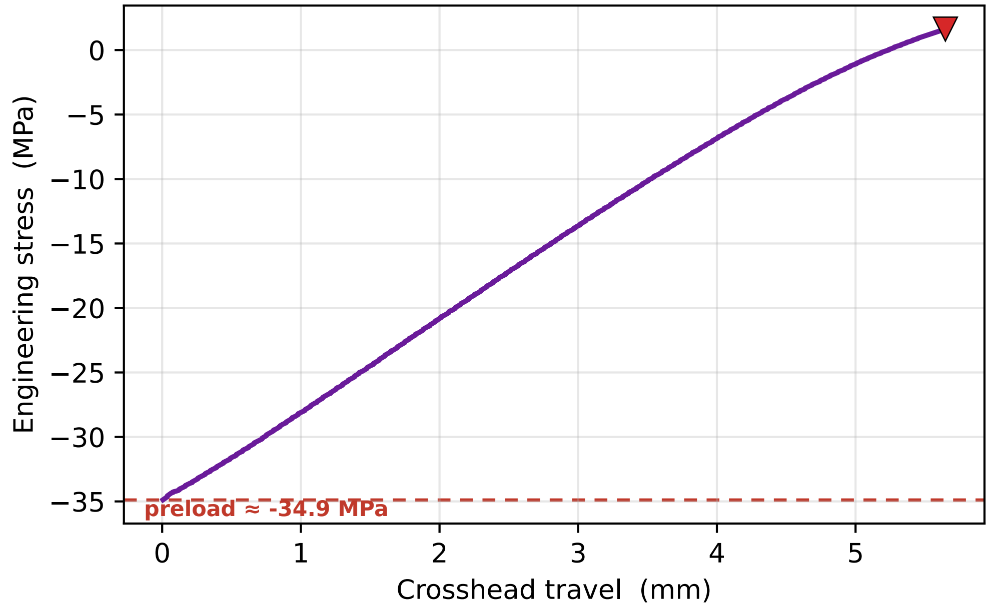


VALIDATION vs REFERENCES

prop	meas	Chacón	E-PLA k
UTS	1.6	**32–60 low**	×35.69
σ_y	-3.8	**30–50 low**	×-15.43
E	1.6	**3–5.5 low**	×1.80
ε_f	2.8 %	(range 3–7 %)	×2.83

⚠ strength outside journal range

Stress vs crosshead



DIC strain vs crosshead — the compliance gap